Voice Input – Functional Workflow

Project: TaskFlow – Intelligent Task Allocation & Tracking System
Parent Module: TaskFlow Assistant (TFA) · **Child Module:** 09.5 Voice Input
Version: 1.0 · **Last Updated:** 31-Oct-2026
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1. Overview

Voice Input (hands-free) lets users talk to TFA instead of typing. Saying *"Hey TF assistant"* opens the TFA Panel and greets the user; the spoken question is then sent like a typed message and TFA's answer is read aloud. A microphone button also lets the user dictate a single command.

2. Purpose / Business Value

- Hands-free access to TFA while the user is busy with other work.
- Faster input for short questions such as "what are my upcoming tasks".
- Privacy: speech is turned into text by the browser's own speech service; TaskFlow keeps no recording.

3. Who Can Access

Role	Can See	Can Use	Access Level (Own / Team / Organization)
Administrator	✅	✅ (Chrome / Edge)	Own
Manager	✅	✅ (Chrome / Edge)	Own
Employee	✅	✅ (Chrome / Edge)	Own
Custom Role	Configurable	Configurable	As granted (`assistant.chat`)

Notes:

- Voice controls are part of the TFA Panel, so they appear only for holders of `assistant.chat`.
- Voice works only in browsers with speech recognition (Chrome, Edge). Typing always works.

4. Screens

#	Screen Name	Menu Path	Purpose	Visible To
1	TFA Panel (voice controls)	Floating **TFA** button, or say "Hey TF assistant"	Talk to TFA and hear answers	Users holding `assistant.chat`, in Chrome or Edge

5. Screen & Functionality Details

5.1 Screen: TFA Panel (voice controls)

Purpose: Talk to TFA instead of typing, and hear the answer read aloud. **Who can see this screen:** Holders of `assistant.chat`, in browsers with speech recognition (Chrome or Edge). **Required permission:** `assistant.chat`

Screen Elements

- Microphone button in the message box (*"Speak a command (Chrome or Edge)"*).
- Listening bar: *Waiting for "Hey TF assistant"*, *Listening... say your command*, *Hands-free is off*, *Voice is paused*, with **Turn on "Hey TF assistant"** / **Turn on hands-free** / **Turn off**.
- Heard text under the TFA button: *Heard: "..."*.
- Voice hint under the floating **TFA** button: *say "Hey TF assistant"*, *Listening... say your command* or *voice is off*.

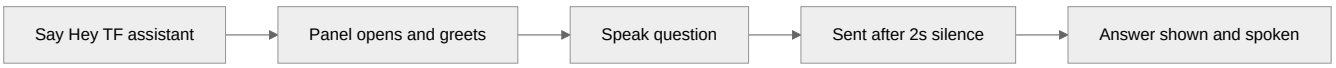
5.2 Functionalities

#	Functionality	Purpose	Who Can See	Who Can Use	Permission Required	Scope
1	Wake phrase	Saying "Hey TF assistant" opens TFA and greets the user by first name	Holders of `assistant.chat`	Holders of `assistant.chat` (Chrome/Edge)	`assistant.chat`	Own
2	Dictate a command	Mic button listens once and fills the message box	Holders of `assistant.chat`	Holders of `assistant.chat` (Chrome/Edge)	`assistant.chat`	Own
3	Read answer aloud	Answers to spoken questions are also spoken (up to 500 characters)	Holders of `assistant.chat`	Holders of `assistant.chat`	`assistant.chat`	Own
4	Turn hands-free on/off	Stop or start listening for the wake phrase	Holders of `assistant.chat`	Holders of `assistant.chat`	`assistant.chat`	Own (remembered in the browser)

5.3 Step-by-Step Workflow – Ask by Voice

1. User says *"Hey TF assistant"*.
2. System opens the panel and says *"Hi {first name}, how can I help you?"*.
3. User speaks a question; the words appear in the message box while speaking.
4. After about 2 seconds of silence, the question is sent like a typed message (the Chat rules in `09.1-Chat.md` apply).
5. TFA's answer is shown and read aloud. The microphone rests while TFA speaks.

Flow Diagram



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5.4 Business Rules & Validations

- Voice works only in browsers with speech recognition (Chrome, Edge). Typing always works.
- Speech is turned into text by the browser's own speech service; TaskFlow keeps no recording.
- Only answers to spoken questions are read aloud; typed questions get text only.
- A deliberate "Turn off" is remembered; an error never turns voice off permanently.

5.5 Messages Shown to the User

Situation	Message
Browser without voice	Voice input is not supported in this browser. Try Chrome or Edge, or type your question.
Microphone blocked	Microphone access was denied. Allow it in your browser to use voice, or type your question.
No microphone	No microphone was found. You can type your question instead.
No internet for voice	Voice needs an internet connection in this browser. You can type your question instead.
Browser blocked auto start	Click anywhere on the page to turn "Hey TF assistant" back on.
Greeting	Hi {first name}, how can I help you?

6. Permission Summary

Permission Key	Description	Administrator	Manager	Employee	Custom Role
`assistant.chat`	Use the TFA Panel, including its voice controls	Granted	Granted	Granted	Configurable

7. Related Modules

- **09.1 Chat** – spoken questions are sent and answered exactly like typed messages.
- **Legal** – the Privacy and Terms pages describe how voice input is handled.

8. Expected Outcome

Users in Chrome or Edge can open TFA by saying "Hey TF assistant", ask their question aloud and hear the answer, with no recording kept by TaskFlow and with typing always available as a fallback.

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